1) Program Charter (Projects A & B)

EPO GenAI Instructor Toolkit POC (Proof of Concept) Program

Purpose

Deliver a credible, demoable **GenAI (Generative Artificial Intelligence)** platform capability (**Project A**) and **Instructor Toolkit Dashboard** workflows (**Project B**) grounded in **OER (Open Educational Resources)** content, with "proof of guardrails," suitable for McGraw Hill internal stakeholder demos and a small instructor pilot.

Timeframe

- **Phase 0 (planning, completed):** 1/26/2026–2/13/2026 — revised Statements of Work (SOWs) finalized
- **Execution:** week of 2/16/2026 through 8/12/2026
- **Formal Sponsor Presentations:** weeks of **3/23**, **5/4**, **6/15**, **8/10**

Scope Summary

- **Project A (Engine):** Content-grounded GenAI Engine using **RAG (Retrieval-Augmented Generation)** with citations, baseline guardrails, templates, application programming interface (API), and basic telemetry.
- **Project B (Dashboard):** Instructor Toolkit Dashboard implementing **two flagship workflows** powered by Project A:
    1. Lecture/Lesson Scaffolding (instructional design aligned)
    2. Assessment Transformation (MCQ → open-ended + rubric)

Constraints (non-negotiables)

- **OER-only** content for the POC unless sponsor explicitly changes this assumption.
- **No platform integration** (Connect/SimNet/ALEKS) in the POC.
- **No WCAG (Web Content Accessibility Guidelines) compliance requirement.**
- **Free vs. Paid Experimentation:** free-tier experimentation allowed early; **must not rely on free-tier** for formal demos/pilot; stable quota-controlled credentials must be in place by **Presentation #1 (week of 3/23)**.
- **No personal student accounts** for shared environments or API keys.
- "Agents" are treated as lightweight **workflow orchestration steps** (not an "agent platform").
- No per-bot virtual machine (VM) / container spin-up architecture in the POC.

Best-practice anchors (quality expectations)

- **Instructional design best practices** for lecture scaffolding (e.g., Backward Design).
- **Assessment design best practices** for transformed questions + rubrics.

- **RAG best practices:** citations/provenance, lightweight evaluation harness (golden set), prompt-injection resilience patterns.

Operating model (discipline with 10 hours/week/student)

- **MVP (Minimum Viable Product) first:** prioritize one strong vertical slice for Presentation #1.
- **Rolling-wave planning:** detailed plan for 4 sprints; beyond that is high-level until Presentation #1.
- **Integration Rule:** enforced every sprint to avoid rework and siloed development.

Governance and cadence

- **Bi-weekly sponsor meeting:** demo + decisions + risk review + next sprint priority adjustments.
- **Change control:** scope additions require sponsor + faculty advisor approval; maintained in a change log.
- **Engineering discipline:** pull requests (PRs) required; code reviews; minimal testing gates; release notes.

Roles (RACI-lite)

- **Sponsor Product Owner:** McGraw Hill sponsor lead (acceptance and prioritization)
- **Faculty Advisor:** Mohammed (scope discipline + academic alignment)
- **Program/Project Manager (PM):** Anirudha (schedule, RAID log—Risks/Assumptions/Issues/Dependencies—meeting orchestration, change log)

Project A Team

- Bao (Tech Lead A / API owner)
- Pragnya (RAG + evaluation/guardrails lead)
- Harsh (safety/testing + API contract tests)
- John (ingestion + telemetry/ops-lite)

Project B Team

- Muthu (Product lead + pilot/test owner)
- Pooja (workflow engineer + exports)
- Rishabh (Integration Bridge + workflow orchestration)
- Sarina (UI/frontend)
- Patricio (Quality assurance (QA) + documentation + checklist operationalization)

2) Project Charter — Project A (Engine)

Mission

Deliver a reusable **content-grounded GenAI Engine** that provides RAG with citations, baseline guardrails, template-driven configurations, basic telemetry, and a stable API for Project B.

Key deliverables (Project A)

- OER ingestion pipeline (chunking, metadata, indexing)
- RAG endpoint (retrieve + generate + citations)
- Template model (2–3 templates max for POC)
- Guardrails baseline ("proof," not production security)
- Engine API v0 + documentation + mock support
- Lightweight evaluation harness (golden set + adversarial prompts)
- Telemetry/usage logging (POC level)
- Integration support for Project B

Success criteria

- Answers grounded in retrieved OER content with citations.
- Out-of-scope refusals and prompt-injection resilience patterns demonstrated in test cases.
- API stability: Project B can integrate in staging; contract tests pass.
- Stable provider/quota path for demos established by week of **3/23**.

Technical guidance (rails, not micromanagement)

- API-first, versioned contract; publish change log for breaking changes.
- Provider-agnostic abstraction but single-provider MVP for demos.
- No per-bot VM/container architecture.
- "Agentic" behavior limited to orchestration steps (retrieve → draft → validate).

3) Project Charter — Project B (Dashboard)

Mission

Deliver an Instructor Toolkit Dashboard that enables instructors to generate, edit, approve, and export outputs for two flagship workflows powered by the Engine, and validate value via a small instructor pilot later in the project.

Key deliverables (Project B)

- Dashboard UI with edit/approve/export flow
- Workflow 1: Lecture/Lesson scaffolding aligned to instructional design best practices
- Workflow 2: Assessment transformation aligned to assessment best practices
- Instructional Design Template Pack + Assessment Quality Checklist
- Pilot protocol + instruments + results summary + refinements (later; after Presentation #2)

Success criteria

- Instructors can complete workflows end-to-end (generate → edit → approve → export).
- Outputs meet checklist expectations in sampling.
- At least one workflow is fully integrated and stable for Presentation #1; both workflows integrated soon after.

Dependency

Project B depends on Project A for:

- RAG + citations/provenance
- guardrail behaviors
- stable API
- stable demo/pilot provisioning (Free then paid experimentation)

4) Sprint 1–4 Rolling Plan (2-week sprints, adjusted for 10 hrs/week)

Key realism tweak:

- **Sprint 2:** require **ONE workflow integrated live** (Lecture scaffolding).
- **Sprint 3 (Presentation #1 week of 3/23):** require **one rock-solid vertical slice**; workflow 2 can be a "preview" (mock or partially live) if needed.
- **Exports:** through Sprint 4, exports are **text/markdown + doc/PDF-friendly** only; PPTX is deferred unless sponsor later elevates it.
- **Ingestion:** start with **1–2 chapters**; expand later.

Sprint summary table

Sprint	Dates (2026)	Goal	Sponsor touchpoint	"Must-hit" outcome
Sprint 1	Feb 16–Feb 27	Converge on tools/stack; lock API contract; mocks + skeletons; OER selection	Bi-weekly demo end of sprint	Dashboard calls Engine mocks; OER chosen; ingestion plan ready; "decision pack" delivered
Sprint 2	Mar 2–Mar 13	RAG v0 live w/ citations; **Lecture workflow integrated live** end-to-end	Bi-weekly demo end of sprint	One workflow runs end-to-end against live Engine in staging (Integration Rule)
Sprint 3	Mar 16–Mar 27	Presentation #1 readiness: **one rock-solid vertical slice**; guardrails + eval snapshot; stable provider	**Formal Presentation #1 week of Mar 23**	Presentation-ready demo; stable credentials/quotas in place; workflow 2 shown as preview if needed
Sprint 4	Mar 30–Apr 10	Integrate workflow 2 fully; improve quality; tighten exports; pilot protocol draft	Bi-weekly demo end of sprint	Both workflows integrated end-to-end in staging with citations and export v1

Sprint 1 detailed plan (Feb 16–Feb 27)

Sprint 1 goals

- Converge on technical approach without micromanaging.
- Enable parallel work: API contract + mock server + dashboard skeleton.
- Choose OER and define ingestion strategy.
- Establish evaluation approach early (lightweight).

Sprint 1 deliverables (with owners and DoD)

Deliverable	DRI (Directly Responsible Individual)	Required collaborators/reviewers	DoD (Definition of Done)
Architecture Decision Record (ADR) Pack v1 *(ADR = a short doc recording a major architecture decision)*	Bao	Pragnya, Harsh, John; review by Rishabh + Anirudha	ADRs approved and in repo: provider plan (Free then paid experimentation), RAG stack choice, app stack choice, deployment-lite plan, key management rules (no personal accounts)
OER selection + ingestion plan	John	Bao, Pragnya; sponsor visibility via Muthu	One OER selected; 1 sample chapter extracted; chunking + metadata plan drafted (chapter/section provenance)
Engine API contract v0 *(API = Application Programming Interface)*	Bao	Rishabh, Harsh, Pooja, Sarina	Contract documented with request/response examples for: lecture scaffolding, assessment transform, citations format, config parameters; change log template created
Mock server running for Engine API	John	Bao, Rishabh, Sarina	Mock service runs locally or in dev; dashboard can call mock and receive predictable structured responses
Dashboard skeleton + workflow wireframes	Sarina	Muthu, Pooja, Rishabh	UI skeleton runs; can submit inputs and display mock output; wireframes for two workflows reviewed by Muthu
Instructional Design framework decision + Template Pack outline	Muthu	Patricio, Pooja; Mohammed review	Framework chosen (e.g., Backward Design); template pack outline includes required inputs/outputs and quality rules

Assessment Quality Checklist draft	Patricio	Muthu, Pooja	Checklist drafted and placed in repo; used as future acceptance aid
Evaluation harness plan *(golden set + adversarial prompts)*	Pragnya	Harsh	Initial test set (10–15 prompts) created; includes citation checks and out-of-scope tests
Contract test harness stub	Harsh	Bao, Rishabh	Basic contract test scaffolding exists (even if against mock); documented how to run

Sprint 1 demo to sponsor (bi-weekly):

- Dashboard calling mock Engine
- OER chosen + ingestion plan
- API contract overview
- Free then paid experimentation provisioning plan and “stable by 3/23” commitment

Sprint 2 detailed plan (Mar 2–Mar 13)

Sprint 2 goals

- Engine RAG v0 live with citations for 1–2 chapters.
- Integrate **one workflow (lecture scaffolding)** end-to-end live in staging.
- Establish baseline guardrails and a measurable evaluation snapshot.

Sprint 2 deliverables

Deliverable	**DRI**	**Required collaborators/reviewers**	**DoD**
Ingestion pipeline v0 + vector index for 1–2 chapters	John	Bao, Pragnya	Script reruns reliably; vector index built; provenance metadata captured
RAG endpoint v0 (retrieve + generate + citations)	Bao	Pragnya	Endpoint returns structured response + citations; at least 10 golden prompts produce cited answers
Guardrails baseline v0 *(course-only + refusal patterns)*	Harsh	Pragnya, Bao	Demonstrated refusals for agreed out-of-scope prompts; injection patterns considered; documented known limitations
Evaluation snapshot v0	Pragnya	Harsh	Golden prompts run; failure cases logged; 1-page summary shared for demo
Workflow 1 integrated live: Lecture scaffolding v0	Pooja	Sarina, Rishabh, Muthu	**Integration Rule met:** dashboard → live Engine → output displayed with citations → edit → export (text/markdown)
Integration checklist + demo script v1	Rishabh	Harsh, Bao	Checklist updated; smoke test steps documented; contract tests pass for workflow 1
UI output formatting + provenance display	Sarina	Pooja, Muthu	Citations visible and readable; structured sections appear per ID template outline

Sprint 2 sponsor demo:

- Live RAG with citations
- Lecture scaffolding end-to-end live
- Guardrails demo
- Evaluation snapshot review

Sprint 3 detailed plan (Mar 16–Mar 27) — Formal Presentation #1 week of Mar 23

Sprint 3 goals (realism-tuned)

- Deliver **one rock-solid vertical slice** for Presentation #1.
- Ensure **stable provisioning** (Free then paid experimentation) for formal demos.
- Workflow 2 can be a **preview** (mocked or partially live) if needed.

Sprint 3 deliverables

Deliverable	DRI	Required collaborators/reviewers	DoD
Stable provider/quota cutover	Bao + Anirudha	Harsh, John	Runs on stable quota-controlled credentials; runbook documented; not dependent on free-tier for presentation
Engine v0 release package	Bao	John, Harsh	API docs updated; version tagged; change log updated; deployment steps documented
Evaluation suite expanded (25–40 prompts)	Pragnya	Harsh	Expanded golden set; adversarial/out-of-scope tests run; summary prepared for presentation
Guardrails hardening for demo	Harsh	Bao, Pragnya	Demonstrable refusal behaviors; injection examples handled; messaging consistent
Vertical slice demo package	Muthu (narrative) + Bao (engine) + Pooja (workflow)	All leads; Mohammed review	Rehearsed demo: workflow 1 end-to-end live, reliable, polished enough; supporting evidence (citations, guardrails, eval snapshot) included
Workflow 2 preview (assessment transform)	Pooja	Patricio, Sarina, Rishabh, Muthu	Demonstrable preview: UI flow and output format aligned to checklist; may call mock engine or partial live endpoint (clearly labeled as preview)

Formal Presentation #1:

- Engine v0 + citations + guardrails proof + evaluation snapshot
- Dashboard demo with **lecture workflow end-to-end live**
- Assessment transform preview (if not fully live yet)
- Confirm readiness path to May 4 milestone

Sprint 4 detailed plan (Mar 30–Apr 10)

Sprint 4 goals

- Fully integrate **workflow 2 live** and stabilize both workflows end-to-end.
- Improve retrieval and formatting based on early feedback.
- Draft pilot protocol (to refine in later sprints before May 4).

Sprint 4 deliverables

Deliverable	DRI	Required collaborators/reviewers	DoD
Workflow 2 integrated live: Assessment transform v1	Pooja	Rishabh, Sarina, Patricio, Muthu; depends on A	**Integration Rule met:** dashboard → live Engine → rubric/rationale output → edit → export (doc/PDF-friendly text)
Assessment checklist v1 operationalized	Patricio	Muthu, Pooja	Checklist applied to sample outputs; issues logged; improvement tasks created
Retrieval quality improvements	Pragnya	John, Bao	Chunking/metadata/prompt improvements implemented; evaluation rerun shows fewer failures
Telemetry/usage logging v1	John	Bao	Basic logs (counts, latency, errors) available; no PII
UI usability improvements + output polish	Sarina	Muthu, Pooja	Inputs simplified; outputs more readable; citations consistent
Integration regression tests + demo script v2	Rishabh	Harsh	Contract tests + smoke tests runnable; demo script updated
Pilot protocol draft v0	Muthu	Patricio	Draft tasks + feedback instrument; cohort ask prepared for sponsor review in Sprint 5

Sprint 4 sponsor demo:

- Both workflows end-to-end live in staging
- Export formats stable (text/markdown/doc/PDF-friendly)
- Pilot protocol draft and recruitment asks previewed